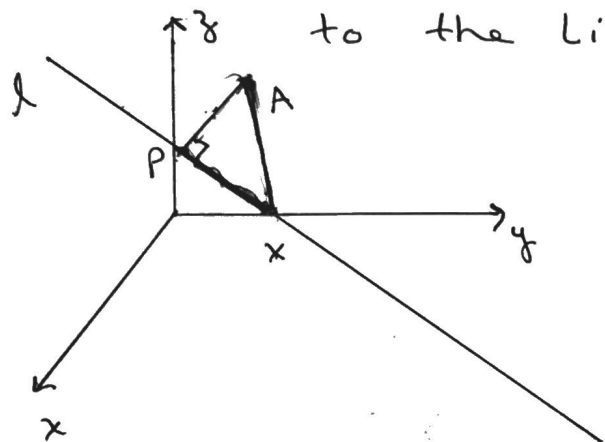


①

## Distance Between a Point and a Line

Example: Find the distance from point  $A(1, 2, 3)$  to the line  $\vec{r} = [0, 1, 5] + t[3, 4, 1]$



① pick a point on the line,  $l$ .

② find the magnitude of the projection of  $\vec{AX}$  onto  $l$ .  
(ie:  $|\vec{PX}|$ )

③ using Pythagorean theorem with hypotenuse length  $|\vec{AX}|$ , find the distance of  $\vec{AP}$ .

→ check if  $A(1, 2, 3)$  is on the line:

$$\left. \begin{array}{l} 1 = 3t \longrightarrow t = \frac{1}{3} \\ 2 = 1 + 4t \longrightarrow t = \frac{1}{4} \\ 3 = 5 + t \longrightarrow t = -2 \end{array} \right\} \Rightarrow A(1, 2, 3) \text{ is not on the line, } l.$$

→ pick a point on the line,  $l$ .

use  $(0, 1, 5)$

(2)

→ find length of  $\vec{PX}$  using proj of  $\vec{AX}$  onto  $\vec{d}$ .

$$|\vec{PX}| = |\text{proj}_{\vec{d}} \vec{AX}| = \frac{|\vec{AX} \cdot \vec{d}|}{|\vec{d}|}$$

$$= \frac{|[-1, -1, 2] \cdot [3, 4, 1]|}{\sqrt{9 + 16 + 1}}$$

$$= \frac{|-3 - 4 + 2|}{\sqrt{26}}$$

$$= \frac{5}{\sqrt{26}}$$

$$\begin{aligned} \rightarrow \text{Length of } \vec{AX} &= \sqrt{(-1)^2 + (-1)^2 + (2)^2} \\ &= \sqrt{6} \end{aligned}$$

→ By Pythagoras,

$$(AP)^2 + (PX)^2 = (AX)^2$$

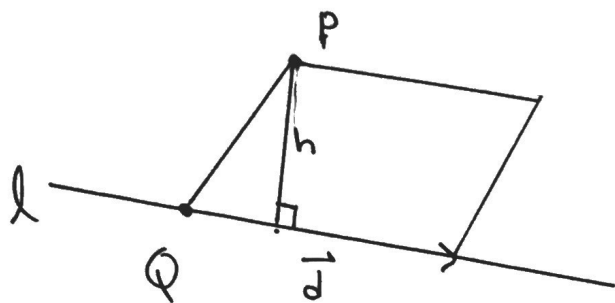
$$(AP)^2 = (\sqrt{6})^2 - \left(\frac{5}{\sqrt{26}}\right)^2$$

$$(AP)^2 = 6 - \frac{25}{26}$$

$$AP \doteq 2.24 \text{ units.}$$

### Shortest distance from a point to a line.

#### Method 2



$$\left. \begin{array}{l} \text{Area of} \\ \text{Parallelogram} \end{array} \right\} = |\vec{PQ} \times \vec{d}| \quad \left\{ \begin{array}{l} \text{Area of} \\ \text{Parallelogram} \end{array} \right. = h \cdot |\vec{d}|$$

$h$  is the shortest distance

$$\therefore h = \frac{\text{Area}}{|\vec{d}|}$$

$$h = \frac{|\vec{PQ} \times \vec{d}|}{|\vec{d}|}$$

## Distance Between 2 Lines

4

- NOTES:
- ① The distance between 2 non-skew lines is 0. (ie: they intersect)
  - ② The shortest distance between skew lines is the length of the common  $\perp$  line.

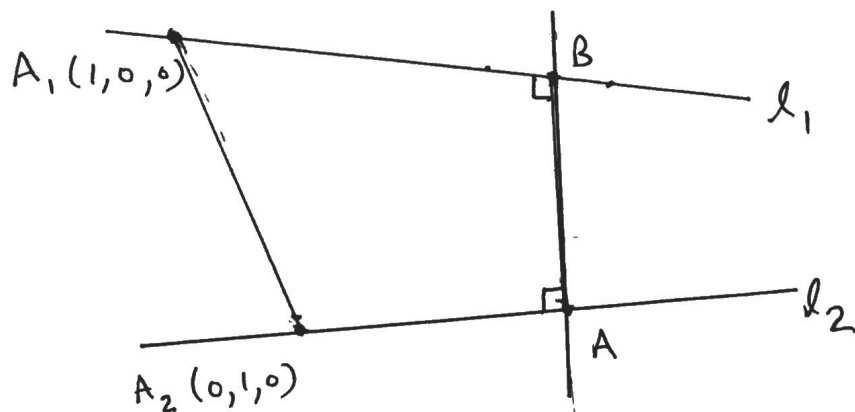
Example: Determine the distance between the 2 skew lines

$$l_1: x-1=y=z \text{ and } l_2: \frac{x}{2} = \frac{y-1}{-1} = z$$

- ① Find the direction vectors and points on the lines.

$$\vec{d}_1 = [1, 1, 1] ; \vec{d}_2 = [2, -1, 1]$$

$$A_1 = (1, 0, 0) ; A_2 = (0, 1, 0)$$



(5)

If we project vector  $\overrightarrow{A_2A_1}$  onto vector  $\overrightarrow{AB}$  and take the magnitude, then that yields the required distance.

$$\therefore \overrightarrow{A_2A_1} = [-1, 1, 0]$$

$$\vec{n} = \vec{d}_1 \times \vec{d}_2$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix}$$

$$\vec{n} = [2, 1, -3]$$

$$\therefore |\overrightarrow{AB}| = \left| \text{proj}_{\vec{n}} \overrightarrow{A_2A_1} \right|$$

$$|\overrightarrow{AB}| = \frac{|\overrightarrow{A_2A_1} \cdot \vec{n}|}{|\vec{n}|}$$

$$|\overrightarrow{AB}| = \frac{|[-1, 1, 0] \cdot [2, 1, -3]|}{\sqrt{(2)^2 + (1)^2 + (-3)^2}}$$

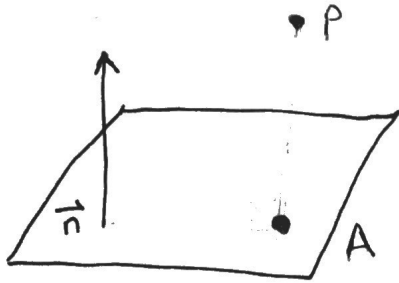
$$|\overrightarrow{AB}| = \frac{1}{\sqrt{14}}$$

$$|\overrightarrow{AB}| \approx 0.27 \text{ units}$$

## Distance from a Point to a Plane.

(6)

ex: Find the distance from the point  $P(-2, 5, 8)$   
to the plane  $2x + 3y - 7z + 5 = 0$



$$\vec{n} = [2, 3, -7]$$

$$\text{let } x = -1 ; y = -1 ,$$

$$\Rightarrow 2(-1) + 3(-1) - 7z + 5 = 0$$

$$\therefore z = 0$$

$\therefore$  The pt. on the plane is  $A(-1, -1, 0)$

$$\vec{AP} = [-1, 6, 8]$$

$$\therefore |\text{proj}_{\vec{n}} \vec{AP}| = \frac{|\vec{AP} \cdot \vec{n}|}{|\vec{n}|}$$

$$= \frac{|[-1, 6, 8] \cdot [2, 3, -7]|}{\sqrt{4+9+49}}$$

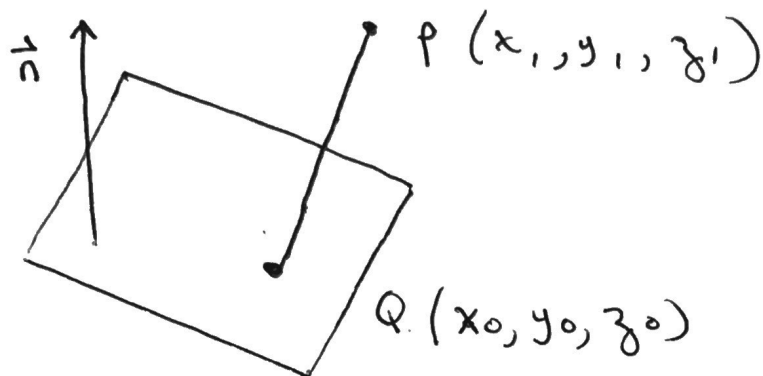
$$= \frac{|-2+18-56|}{\sqrt{62}}$$

$$= \frac{40}{\sqrt{62}}$$

$$\approx 5.08 \text{ units}$$

## Distance from a point to a Plane.

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$$\vec{QP} = [x_1 - x_0, y_1 - y_0, z_1 - z_0]$$

Equation of Plane is  $Ax + By + Cz + D = 0$

point  $Q(x_0, y_0, z_0)$  is on the plane.

$$\therefore A(x_0) + B(y_0) + C(z_0) + D = 0$$

$$\Rightarrow D = -Ax_0 - By_0 - Cz_0$$

$$\text{distance} = |\text{proj}_{\vec{n}} \vec{QP}|$$

$$\text{distance} = \frac{|\vec{QP} \cdot \vec{n}|}{|\vec{n}|}$$

$$\text{distance} = \frac{[x_1 - x_0, y_1 - y_0, z_1 - z_0] \cdot [A, B, C]}{\sqrt{A^2 + B^2 + C^2}}$$

$$\text{distance} = \frac{|A(x_1 - x_0) + B(y_1 - y_0) + C(z_1 - z_0)|}{\sqrt{A^2 + B^2 + C^2}}$$

(8)

$$\text{distance} = \frac{|Ax_1 - Ax_0 + By_1 - By_0 + Cz_1 - Cz_0|}{\sqrt{A^2 + B^2 + C^2}}$$

$$\text{distance} = \frac{|Ax_1 + By_1 + Cz_1 - Ax_0 - By_0 - Cz_0|}{\sqrt{A^2 + B^2 + C^2}}$$

$$\text{distance} = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$



## Distance Between a Line & a plane.

⑨

ex: find the distance from line

$$l: \vec{r} = [2, 0, 1] + t[1, 4, 1]$$

$$\pi: 2x - y + 2z - 4 = 0$$

$\therefore$  parallel.

Solution: note:  $\vec{d} \cdot \vec{n} = [1, 4, 1] \cdot [2, -1, 2] = 0$

$$d = \frac{|\vec{n} \cdot \vec{PQ}|}{|\vec{n}|}$$

$$\vec{n} = [2, -1, 2]$$

$$P = (2, 0, 1)$$

$$Q: \text{Let } x=1, y=1$$

$$2(1) - (1) + 2z - 4 = 0$$

$$2z = 3$$

$$z = \frac{3}{2}$$

$$\therefore Q = (1, 1, \frac{3}{2})$$

$$\begin{aligned} \vec{PQ} &= (1, 1, \frac{3}{2}) - (2, 0, 1) \\ &= [-1, 1, \frac{1}{2}] \end{aligned}$$

$$d = \frac{|[2, -1, 2] \cdot [-1, 1, \frac{1}{2}]|}{\sqrt{(2)^2 + (-1)^2 + (2)^2}}$$

$$d = \frac{|-2 - 1 + 1|}{\sqrt{4 + 1 + 4}}$$

$$d = \frac{2}{\sqrt{9}}$$

$$d = \frac{2}{3} \text{ units.}$$

## Distance Between 2 Planes

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ex: Find the distance between the planes

$$\pi_1: 2x - y - z - 1 = 0$$

$$\pi_2: 2x - y - z - 4 = 0$$

Solution: note:  $\vec{n}_1 = \vec{n}_2 \Rightarrow$  parallel

Find a point on  $\pi_1$ : let  $x$  and  $y = 0$

$$\Rightarrow z = -1$$

$$\therefore P = (0, 0, -1)$$

Find a point on  $\pi_2$ : let  $x$  and  $y = 0$

$$\Rightarrow z = -4$$

$$Q = (0, 0, -4)$$

$$\vec{PQ} = [0, 0, -3]$$

$$d = \frac{|[2, -1, -1] \cdot [0, 0, -3]|}{\sqrt{(2)^2 + (-1)^2 + (-1)^2}}$$

$$d = \frac{3}{\sqrt{4+1+1}} = \frac{3}{\sqrt{6}} \doteq 1.22 \text{ units}$$